Laborator 8

1.

```
TSP(G, L)
    V' = ∅
    for i = 1, ---, n
        u = choice(V)
        if u ∈ V'
            fail
        V' = V' ∪ {u}
    suma = 0
    V' = V' ∪ {v'[1]}
    for i = 1, ---, n
        suma = suma + cost[v'[i]][v'[i+1]]
    if suma <= L
        success
    fail
```

Generare $O(N^2)$

$O(N)$

$O(1)$

Testare

Complexitatea temporală: $O(n^2+n+1) = O(n^2)$

⇒ TSP aparține NP.

(2.)

```
B(G, NC) → O(N^2)
    for i=0,...,n-1          → O(N)      } O(N)
        c[i] = choice(1,...,NC) → O(1)   }
    for each (x,y) ∈ E        → O(N^2)   } O(N^2)
        if c[x] = c[y]        → O(1)     }
            fail              → O(1)
    success                   → O(1)
```

```
A(G, k)
    for i=1,...,N             → O(N)     } O(N^3)
        if B(G,i)             → O(N^2)   }
            break             → O(1)     }
    if i=k                    → O(1)
        success               → O(1)
                              → O(1)
    fail
```

Complexitatea este $O(N^3)$.